Namit Anand

Curriculum Vitæ

NASA Quantum AI Lab
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Bio

I'm a Postdoctoral Researcher (→ Staff Scientist in 2024!) working in quantum information theory and quantum computation at the Quantum AI Lab (QuAIL) at NASA Ames Research Center and KBR. I received my PhD in Physics from the University of Southern California (USC) under the (wonderful) supervision of Paolo Zanardi. My coadvisors were Todd Brun and Aaron Lauda. Before that, I received my Integrated Master of Science (BS-MS) in Physics from the National Institute of Science Education and Research (NISER), Bhubaneswar.

Research interests

Application of quantum information-theoretic tools to many-body theory (specifically, information scrambling, quantum chaos, and many-body localization), quantum resource theories (specifically, quantum coherence and entanglement), open quantum systems and non-Markovian processes, quantifying the role of coherence and entanglement in quantum algorithms, error mitigation for NISQ-devices, non-universal models of quantum computation (Clifford group, matchgates, etc.)

Employment

2024 – present **Staff Scientist**, *NASA Quantum AI Lab (QuAIL) and KBR*, Mountain View, CA, USA.

2022 – 2023 **Postdoctoral Researcher**, *NASA Quantum AI Lab (QuAIL) and KBR*, Mountain View, CA, USA.

Education

2017 – 2022 **Doctor of Philosophy, Physics**, *University of Southern California*, Los Angeles, CA, USA.
Advisor: Paolo Zanardi. *Co-advisors*: Todd A. Brun and Aaron Lauda.
Thesis: Quantum information-theoretic aspects of chaos, localization, and scrambling

2016 – 2017 **Visiting Student Researcher, Physics**, *Harish-Chandra Research Institute*, Prayagraj, India.

2011 – 2016 **Integrated Master of Science (BSc +MSc), Physics**, *National Institute of Science Education and Research*, Bhubaneswar, India.

Publications & Preprints

1. **Simulation of quantum optics by coherent state decomposition**, *Jeffrey Marshall and Namit Anand*, arXiv:2305.17099 (2023).

2. **Scrambling and operator entanglement in local non-Hermitian quantum systems**, *Brian Barch, Namit Anand, Jeffrey Marshall, Eleanor Rieffel, Paolo Zanardi*, arXiv:2305.12054 (2023).
3. **Scrambling of Algebras in Open Quantum Systems**, *Faidon Andreadakis, Namit Anand and Paolo Zanardi*, Phys. Rev. A **107**, 042217 (2023), (Editors' suggestion).
4. **BROTOCs and Quantum Information Scrambling at Finite Temperature**, *Namit Anand and Paolo Zanardi*, Quantum **6**, 746 (2022).
5. **Quantum coherence as a signature of chaos**, *Namit Anand, Georgios Styliaris, Meenu Kumari, and Paolo Zanardi*, Phys. Rev. Research **3**, 023214 (2021).
6. **Information Scrambling and Chaos in Open Quantum Systems**, *Paolo Zanardi and Namit Anand*, Phys. Rev. A **103**, 062214 (2021).
7. **Information Scrambling over Bipartitions: Equilibration, Entropy Production, and Typicality**, *Georgios Styliaris, Namit Anand, and Paolo Zanardi*, Phys. Rev. Lett. **126**, 030601 (2021).
8. **Quantum coherence and the localization transition**, *Georgios Styliaris, Namit Anand, Lorenzo Campos Venuti, and Paolo Zanardi*, Phys. Rev. B **100**, 224204 (2019).
9. **Quantifying non-Markovianity: a quantum resource-theoretic approach**, *Namit Anand and Todd A. Brun*, arXiv:1903.03880 [quant-ph] (2019), (submitted to Physical Review Letters).
10. **Demonstration of fidelity improvement using dynamical decoupling with superconducting qubits**, *Bibek Pokharel, Namit Anand, Benjamin Fortman, and Daniel A. Lidar*, Phys. Rev. Lett. **121**, 220502 (2018).
    – Popular summary of our work in Phys.org, ScienceDaily, and the Scientific American.
    – Featured on the Rigetti website.
11. **Asymmetry and coherence weight of quantum states**, *Kaifeng Bu, Namit Anand, and Uttam Singh*, Phys. Rev. A **97**, 032342 (2018).
12. **Coherence and Entanglement Monogamy in the Discrete Analogue of Analog Grover Search**, *Namit Anand and Arun K. Pati*, arXiv:1611.04542 [quant-ph] (2016).
13. **Comment on "Limitations on the superposition principle: superselection rules in non-relativistic quantum mechanics"**, *Namit Anand*, Eur. J. Phys. **37** 048003 (2016).
14. **Do quantum strategies always win?**, *Namit Anand and Colin Benjamin*, Quantum Information Processing **14** (11), 4027-4038 (2015).

⋆ [Google Scholar] and [arXiv].

Academic Service

Referee QIP 2023, Physical Review X, Physical Review Letters, Physical Review Research, Physical Review B, Physical Review E

Conference Organizer YQIS 2021

Invited Seminars

1. **Information-theoretic aspects of scrambling and chaos**, *IIT Madras, India*, Feb. 11, 2022, Arul Lakshminarayan's group.
2. **Quantum information-theoretic aspects of scrambling, localization, and chaos**, *IIT Kanpur, India*, Aug. 26, 2022, Young Quantum Condensed Matter (YQCM) seminar series.
3. **Quantum information-theoretic aspects of scrambling, localization, and chaos**, *UC San Diego, USA*, Oct. 4, 2022, CMT Journal Club.
4. **Random unitaries, quantum chaos, and all that**, *University of New South Wales, Australia*, May 25, 2023, Andrea Morello's group.
5. **Quantum information-theoretic aspects of scrambling, localization, and chaos**, *University of Luxembourg & Donostia International Physics Center, Spain*, June 29, 2023, Adolfo del Campo and Aurelia Chenu's group.

Teaching Experience

(Fall '17 & Spring '18) Graduate Teaching Assistant for Physics 135aL: Physics for the Life Sciences. Student Evaluation Rating: 5.0/5.0

Awards and Fellowships

2022 Robust Quantum Simulation (RQS) Postdoctoral Fellowship (Declined)

Spring 2022 Feynman Quantum Academy Internship at Quantum Artificial Intelligence Laboratory (QuAIL), NASA Ames via KBR.

Summer 2021 Recipient of the USC Gold Family Fellowship.

Award: $5,000 (USD)

Summer 2018 Recipient of the Quantum computing fellowship at the Los Alamos National Laboratory.

Award: $12,200 (USD)

2017 – 2022 University of Southern California, Department of Physics & Astronomy.

Award: $27,500 (USD) annually; full financial support through teaching assistantship and one year of fellowship for completion of the Ph.D. program.

2017 Graduate fellowships declined for USC: University of Maryland; Dartmouth College; Université libre de Bruxelles; Scuola Normale Superiore di Pisa.

Summer 2015 One of the five summer fellows (internationally) for the fully funded I.I.A.S.S. Summer internship program, Italy.

Award: €3,000.

2011 – 2016 Recipient of Innovation in Science Pursuit for Inspired Research (INSPIRE) scholarship by the Department of Science and Technology, Government of India from 2011 - 2016.

Award: ₹80,000/year for 5 years.

Computational Skills

Programming Languages	C++, Python
Specialized Software	Mathematica and MATLAB
Markup Languages	LaTeX, HTML, HTML5, PHP
Software adept	Adobe Suite (Photoshop, Audition) and Game Development (UDK)
Numerical packages	*Lapack* for C++ & Python, MATLAB on HPC, and CVX for convex optimization in MATLAB
Quantum simulations	Simulating quantum error-correction on the IBM Quantum Experience

Professional References

Paolo Zanardi

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Todd A. Brun

Professor of Electrical Engineering-Systems, Computer Science, and Physics and Astronomy
University of Southern California
Email: tbrun@usc.edu
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Daniel A. Lidar

(*Viterbi Professor of Engineering*)
Professor of Electrical Engineering-Systems, Chemistry, and Physics and Astronomy
University of Southern California
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Arun K. Pati

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(Last updated: August 15, 2023)